MAST32001 Computational Statistics

Course Final Assignment, WS 2026

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Task 1: Time series bootstrap and the ladle estimator

Part A: Whitening and autocovariance matrices

Conventions and implementation

The observations are stored as the rows of an $n \times p$ matrix X. Each variable is centred using its full-sample column mean,

$$X_c = X - \mathbf{1}_n \bar{x}^\mathsf{T}, \qquad S_0 = \frac{1}{n-1} X_c^\mathsf{T} X_c.$$

If $S_0 = VDV^\mathsf{T}$, the symmetric whitening matrix and whitened data are

$$W = S_0^{-1/2} = VD^{-1/2}V^\mathsf{T}, \qquad Y = X_c W.$$

Eigenvalues below 10^{-10} times the largest eigenvalue are treated as numerically zero because their inverse square roots would be unstable. No additional marginal scaling is applied because scaling is already included in W.

For lag $h \geq 1$, we estimate

$$\widehat{C}_h = \frac{1}{n-h} \sum_{t=1}^{n-h} y_t y_{t+h}^\mathsf{T}.$$

The mean is not re-estimated within each lag window. Row t is paired with row $t+h$, and normalization uses the $n-h$ available pairs. The matrices used for joint diagonalization are symmetrized as $\widehat{C}_h^{(s)} = (\widehat{C}_h + \widehat{C}_h^\mathsf{T})/2$.

Purpose and numerical verification

Whitening removes contemporaneous scale and cross-covariance, so the lag-zero sample covariance becomes I_p. In the second-order source-separation model, this changes the unknown mixing matrix into an orthogonal matrix, allowing the remaining directions to be estimated by orthogonal joint diagonalization.

Whitening does not make the observations serially independent; non-zero-lag dependence remains in the autocovariance matrices and provides the information used to identify the signal directions.

For the supplied data, $n = 1000$ and $p = 10$. The numerical checks gave

$$\max_j |\bar{y}_j| = 6.49 \times 10^{-14}, \qquad \max_{j,k} \left| \left\{ \widehat{\mathrm{Cov}}(Y) - I_{10} \right\}_{jk} \right| = 3.55 \times 10^{-15}.$$

The whitened variances ranged from 0.999999999999997 to 1, and the largest absolute off-diagonal covariance was 2.58×10^{-15}. These values confirm the expected whitening effect to numerical precision. The non-zero Frobenius norms of the lagged matrices, including 0.3653 at lag 1 and 0.5575 at lag 2, also show that whitening does not force non-zero-lag covariance estimates to zero.

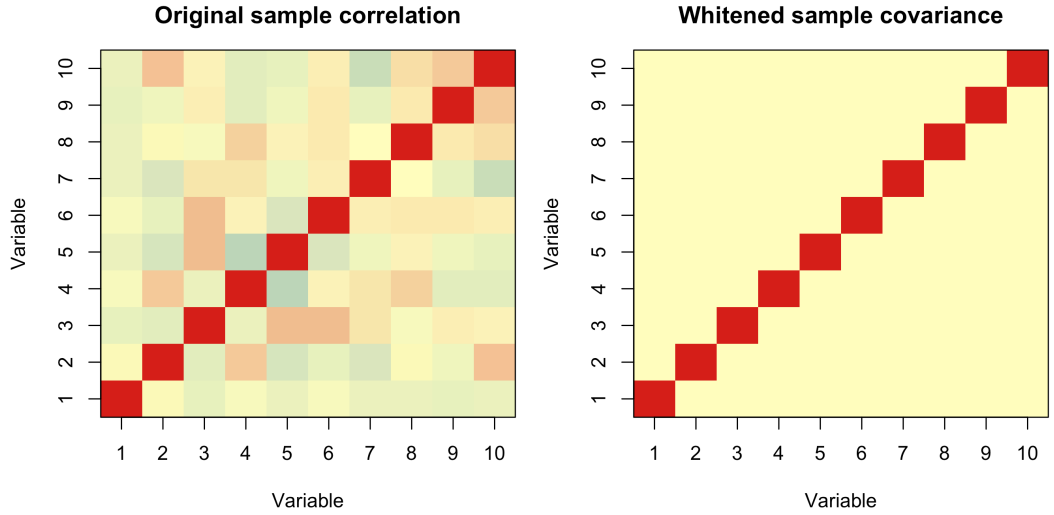


Figure 1: Original contemporaneous correlation and covariance after whitening.

Part B: Bootstrap for dependent observations

Fixed-block bootstrap

For block length L, a starting position is sampled uniformly with replacement from $\{1, \ldots, n - L + 1\}$. The corresponding L consecutive multivariate rows are retained in their original order. This is repeated $\lceil n/L \rceil$ times, after which the blocks are concatenated and truncated to n rows. The tuning parameter L determines how much local dependence can be retained within each block.

Stationary bootstrap

The stationary bootstrap begins at an index sampled uniformly from $\{1, \ldots, n\}$. At every subsequent position, a new uniform index is selected with probability $r = 1/L$; otherwise, the previous index is advanced by one, with circular wrap-around from n to 1. Consequently, each block length K has the geometric distribution

$$\Pr(K = k) = r(1 - r)^{k-1}, \qquad E(K) = \frac{1}{r} = L.$$

Unlike the fixed-block method, the stationary bootstrap uses random block lengths. In 100 generated samples, target expected lengths 10, 20 and 40 gave mean observed lengths 9.94, 20.20 and 40.94, respectively.

Why IID resampling is unsuitable

Resampling individual rows independently destroys temporal ordering and makes non-zero-lag autocovariances approximately zero. Such samples do not represent the variability of statistics calculated from the original dependent process. Both block procedures instead resample consecutive multivariate observations, preserving local serial and cross-series dependence within blocks.

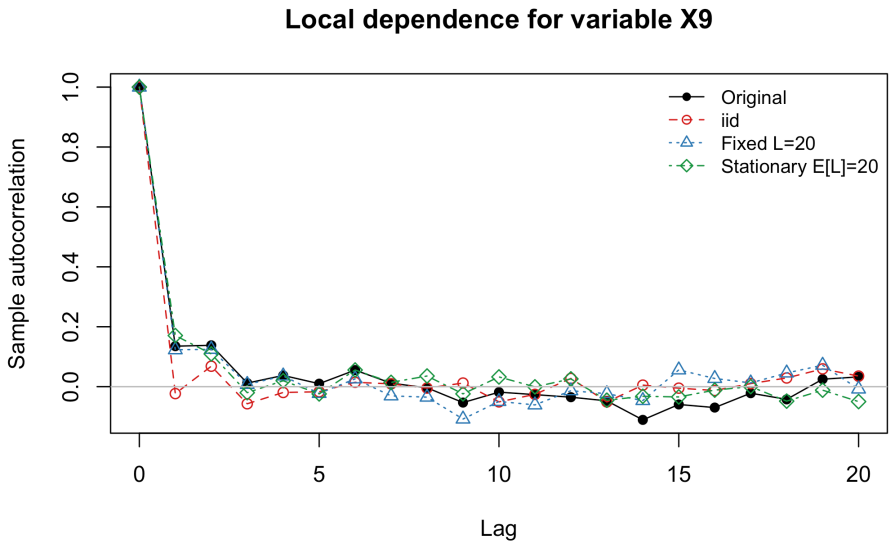


Figure 2: Example ACF comparison for the variable with the largest absolute lag-1 sample ACF.

Effect of block length

Dependence preservation was evaluated using the mean absolute difference between the original and bootstrap sample autocorrelations, averaged over all 10 variables and lags 1–20. Each row of Table 1 summarizes 100 bootstrap samples.

Table 1: Mean absolute ACF error; smaller values indicate better local-dependence preservation.

Method	Mean error	Standard deviation
IID	0.0460	0.0017
Fixed, $L = 10$	0.0381	0.0021
Fixed, $L = 20$	0.0320	0.0019
Fixed, $L = 40$	0.0270	0.0025
Stationary, $E(K) = 10$	0.0336	0.0020
Stationary, $E(K) = 20$	0.0295	0.0024
Stationary, $E(K) = 40$	0.0269	0.0027

IID resampling has the largest error. Longer blocks preserve the observed ACF more closely, although they provide fewer effectively independent blocks and slightly greater Monte Carlo variability. Block length is therefore a tuning choice that balances dependence preservation against the number of resampled blocks.

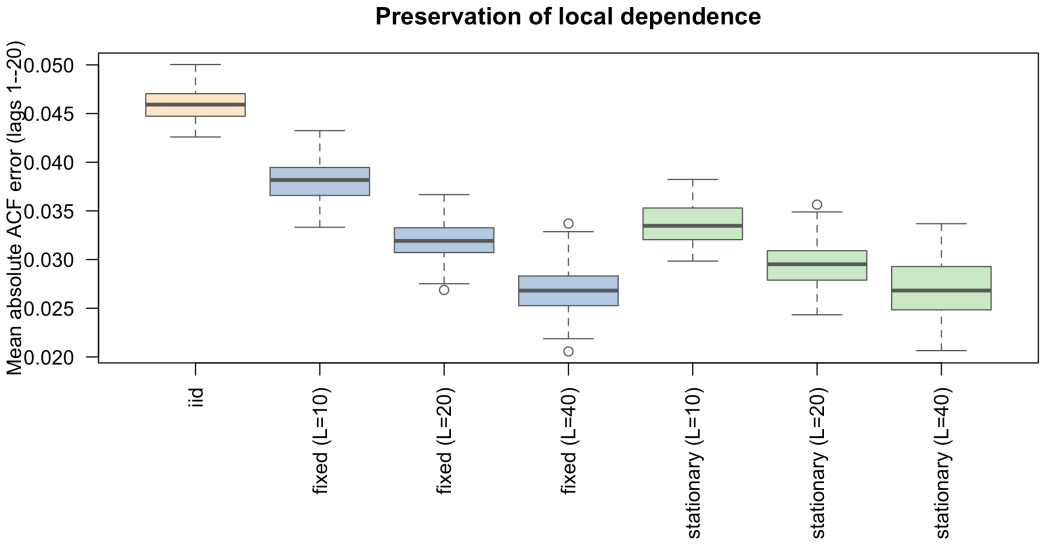


Figure 3: Distribution of ACF error over 100 bootstrap samples for each method and block length.

Part C: The ladle quantities

Eigenvalue-based quantity ϕ

For the lag set $\mathcal{T}$, an orthogonal matrix U approximately jointly diagonalizes the whitened autocovariance matrices. The component strengths are

$$s_j = \sum_{h \in \mathcal{T}} \left[\operatorname{diag}\left(U^{\mathsf{T}} \widehat{C}_h^{(s)} U \right)_j \right]^2,$$

ordered so that $s_0 \geq \cdots \geq s_{p-1}$. The normalized scree contribution is

$$\phi(k) = \frac{s_k}{1 + \sum_{\ell=0}^{p-1} s_\ell}, \qquad k = 0, \ldots, p-1.$$

Large values correspond to directions with serial dependence, while noise directions should have strengths close to zero. When several lags are used, U is obtained with `JADE::rjd`; the strength calculation and ordering are implemented directly.

Bootstrap-variability quantity f

Let $\widehat{B}_k$ and $B^*_{k,b}$ contain the first k ordered directions from the original sample and bootstrap sample b. Their subspace instability is measured by

$$f_0(k) = \frac{1}{B}\sum_{b=1}^{B}\left\{1 - \left|\det\left(\widehat{B}_k^{\intercal} B_{k,b}^{*}\right)\right|\right\}, \quad k = 1, \ldots, p-1, \qquad f_0(0) = 0,$$

and normalized as

$$f(k) = \frac{f_0(k)}{1 + \sum_{\ell=0}^{p-1} f_0(\ell)}.$$

The absolute determinant measures agreement between the two k-dimensional subspaces and is invariant to sign changes of individual directions. Stable signal subspaces have small $f(k)$, while variability increases when noise directions enter the estimated subspace.

Combination and dimension estimate

The ladle criterion combines the two sources of information:

$$g(k) = f(k) + \phi(k), \qquad \hat{q} = \underset{k \in \{0,\ldots,p-1\}}{\arg\min}\; g(k).$$

The scree contribution decreases as weak or noise directions are included, whereas the bootstrap contribution increases when the estimated subspace becomes unstable. Their sum balances signal strength and subspace stability.

Simulation check

The implementation was checked using Setting 1 of Nordhausen and Virta: five Gaussian AR/MA signal series and five Gaussian white-noise series, giving $p = 10$ and true dimension $q = 5$. The check used $n = 1000$, lags 1–12, 100 bootstrap samples and block length 20. Both the fixed-block and stationary bootstrap versions returned $\hat{q} = 5$.

The largest whitening identity error was 3.22×10^{-15}, the largest orthogonality error in $U^{\intercal}U$ was 2.22×10^{-15}, and the joint off-diagonal squared-energy ratio was 0.0718. These checks verify the whitening, orthogonal directions, approximate joint diagonalization and recovery of the known signal dimension.

Part D: Simulation study

Simulation setting and study design

The purpose of the simulation is to assess whether the implementation recovers a known signal dimension, and how its performance changes with sample size, bootstrap method and block length. We use Setting 1 of Nordhausen and Virta, with $p = 10$ components and true signal dimension $q = 5$. The model has identity mixing, $x_t = z_t$, where the first five components are independent Gaussian AR/MA processes and the remaining five are independent Gaussian white-noise series. The signal parameters are listed in Table 2, using the convention

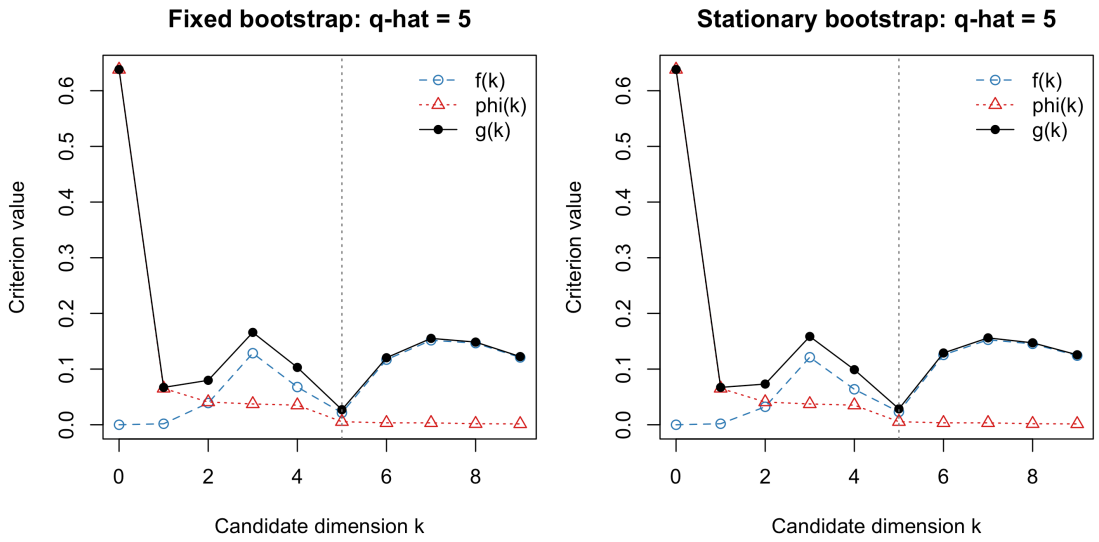


Figure 4: The implemented f, ϕ and g on Setting 1 data. Both bootstrap versions attain their minimum at the true dimension $q = 5$.

$$z_{j,t} = \sum_{\ell=1}^{r_j} a_{j,\ell} z_{j,t-\ell} + \varepsilon_{j,t} + \sum_{\ell=1}^{s_j} b_{j,\ell} \varepsilon_{j,t-\ell}, \qquad \varepsilon_{j,t} \overset{\text{iid}}{\sim} N(0,1).$$

Innovations are independent across components. The five noise series also have independent $N(0,1)$ observations. As in the implementation used in Part C, each generated column is centred and scaled to unit sample variance before the whitening step.

Table 2: Signal processes in Setting 1. A dash denotes an absent AR or MA part.

Component	AR coefficients	MA coefficients
1	(0.6)	$(0, -0.4)$
2	$(0.4, 0.1, 0.3)$	$(0.2, 0.4)$
3	(0.4)	–
4	–	$(0.5, 0.3, 0.1, 0.2)$
5	$(0.1, 0.1, 0.3)$	–

We consider $n \in \{200, 500, 1000\}$, the SOBI lag set $\mathcal{T} = \{1, \ldots, 12\}$, and both fixed-block and stationary bootstrap methods. For each method, $L \in \{10, 20, 40\}$ denotes the fixed or expected block length, respectively. These choices match the corresponding settings in the paper. This study focuses on SOBI, extending the implementation from Parts A–C; it does not repeat the paper's separate AMUSE comparison.

For each sample size, $R = 100$ independent datasets are generated. Each dataset is analysed under all six method–length combinations, giving 1800 ladle fits. Each fit uses $B = 100$ bootstrap samples. The paper uses 2000 simulation repetitions and 200 bootstrap samples per fit; our smaller experiment is intended to reproduce the main qualitative behaviour while keeping the computational cost manageable.

Here R measures variation between independently generated datasets, whereas B controls the Monte Carlo approximation within a single ladle fit.

Computational procedure and tuning choices

For each generated dataset, the calculation proceeds as follows.

1. Centre the observations and whiten them using the symmetric inverse square root of the sample covariance matrix, following the conventions in Part A. Compute the symmetrized autocovariance matrices at lags $1, \ldots, 12$.

2. Apply `JADE::rjd` to obtain the orthogonal joint diagonalizer. Order its directions by decreasing sums of squared diagonal autocovariances. These strengths determine the scree contribution $\phi(k)$ from Part C.

3. Generate B multivariate bootstrap samples using either fixed blocks of length L or stationary blocks with restart probability $1/L$, as in Politis and Romano. For every bootstrap sample, repeat centring, whitening, autocovariance estimation, joint diagonalization and ordering. Thus the bootstrap includes uncertainty in the preprocessing as well as in the estimated directions.

4. Use the determinant-based comparison of the original and bootstrap leading subspaces to compute $f(k)$. Form

$$g(k) = f(k) + \phi(k), \qquad \hat{q} = \operatorname*{arg\,min}_{k=0,\ldots,9} g(k),$$

   and record $\hat{q}$.

Short blocks contain more boundaries and consequently break more lagged pairs. In particular, a fixed block of length 10 cannot preserve a lag-12 pair within a single block. Larger blocks preserve longer stretches of dependence but provide fewer independently resampled blocks; at $n = 200$, $L = 40$ gives only five fixed blocks. The three lengths therefore explore a relevant trade-off rather than assuming that the longest block must always be best.

Explicit seeds are assigned to the generated datasets and bootstrap fits. Sharing each dataset across the six settings permits paired comparisons of the methods, but these six fits are not six independent simulation repetitions. The joint-diagonalization tolerance is 10^{-6}. Its iteration limit is increased from 1000 to 10000 because a few preliminary bootstrap calculations reached the smaller limit; the tolerance is unchanged. All 1800 final fits completed. The maximum whitening identity error was 2.76×10^{-14}, and the maximum direction-orthogonality error was 8.22×10^{-15}.

Monte Carlo summaries and results

For a fixed sample size, method and block length, let $I_r = \mathbf{1}\{\hat{q}_r = 5\}$. The estimated correct-selection probability and its Monte Carlo standard error are

$$\widehat{P} = \frac{1}{R}\sum_{r=1}^{R} I_r, \qquad \widehat{\mathrm{MCSE}}(\widehat{P}) = \sqrt{\frac{\widehat{P}(1-\widehat{P})}{R}}.$$

With $R = 100$, the largest possible MCSE is 0.05. We also report 95% Wilson intervals, which remain informative when the observed success rate is 0 or 1. The complete dimension distribution distinguishes underestimation from overestimation instead of reducing performance to the mean estimated dimension.

Table 3: Percentage of repetitions selecting the true dimension $q = 5$. Each entry is based on 100 independent datasets and $B = 100$ bootstrap samples per fit.

Method	L	$n = 200$	$n = 500$	$n = 1000$
Fixed	10	0	36	99
Fixed	20	0	37	99
Fixed	40	1	38	99
Stationary	10	0	34	99
Stationary	20	1	40	99
Stationary	40	1	45	99

Table 3 and Figure 5 show that sample size has the clearest effect. At $n = 200$, correct selection occurs in only 0–1% of repetitions. Both underestimation and overestimation are common: their rates range from 58–77% and 22–42%, respectively. At $n = 500$, recovery improves to 34–45%, with most remaining errors being underestimates. At $n = 1000$, every setting selects $q = 5$ in 99 repetitions and $q = 4$ in one. The corresponding 95% Wilson interval for the recovery rate is $[94.6\%, 99.8\%]$, so the experiment shows high but not perfectly known population performance.

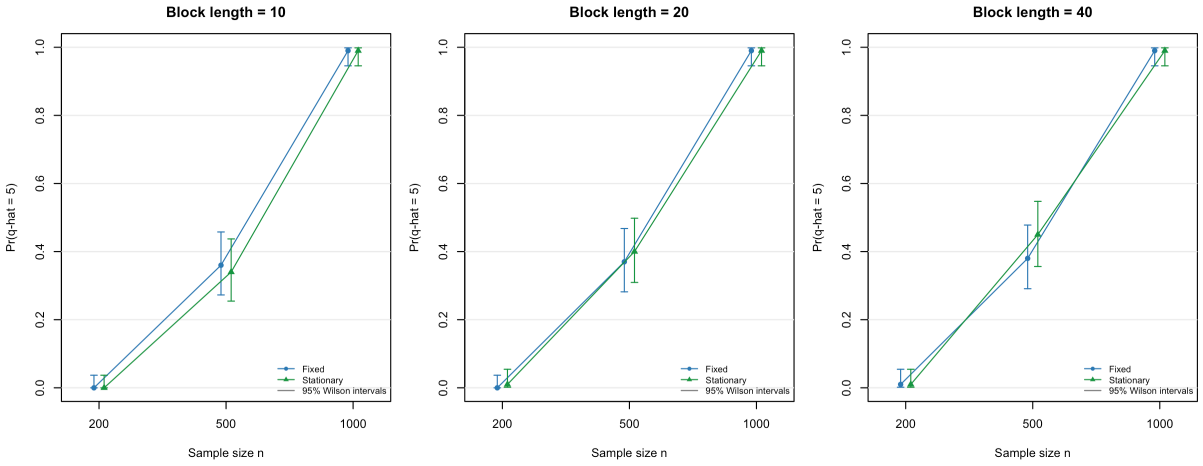


Figure 5: Correct-selection probabilities with 95% Wilson Monte Carlo intervals. Sample sizes are displayed as three discrete design points.

The difference between bootstrap methods is smaller than the sample-size effect. For example, at $n = 500$ the stationary-minus-fixed differences in recovery are -2, 3 and 7 percentage points for $L =$

$10, 20, 40$, with paired MCSEs of 3.48, 3.32 and 3.83 percentage points. These standard errors are computed from the within-dataset differences $I_{r,S} - I_{r,F}$, accounting for the shared datasets. The modest advantage of longer stationary blocks at $n = 500$ should therefore not be treated as decisive evidence that one method is universally superior. The methods give identical dimension selections in this experiment at $n = 1000$.

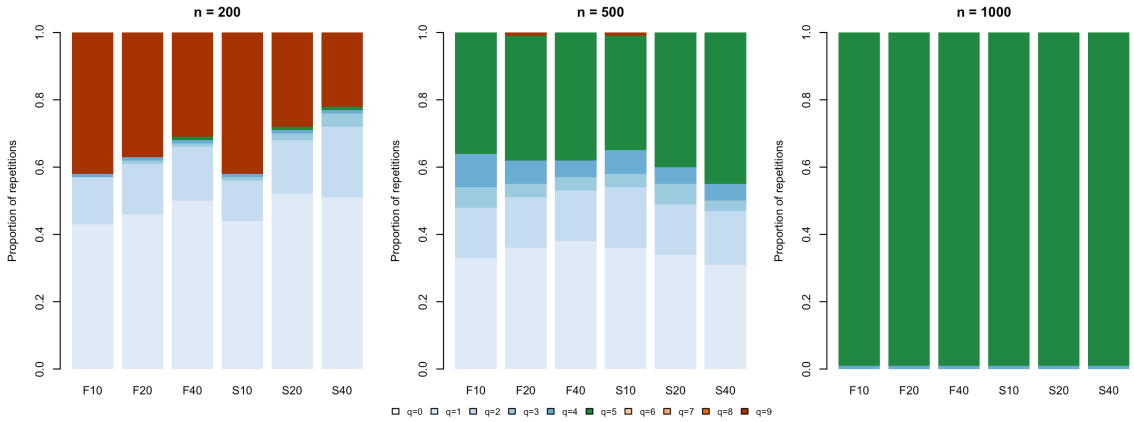


Figure 6: Distribution of estimated dimensions in Setting 1. F and S denote fixed-block and stationary bootstrap; the number denotes L. The true dimension is 5, shown in green.

The broad distributions for the shorter series in Figure 6 demonstrate that their poor recovery is not merely a small bias around $q = 5$. In contrast, the long-series distributions concentrate at the true dimension. This reproduces the main qualitative finding of Nordhausen and Virta: SOBI-based ladle estimation becomes reliable when the series are sufficiently long, while short series can be inadequate. Our results provide numerical validation under Setting 1, rather than a proof of consistency or a precise replication of every result in the paper. They also describe the estimator at the finite bootstrap budget $B = 100$.

Part E: Application to the course data

Application procedure and block-length selection

The supplied `timeseries.dat` contains $n = 1000$ observations of $p = 10$ variables. We apply the same whitening, lag set $\mathcal{T} = \{1, \dots, 12\}$, joint diagonalization and ladle calculations used in Part D. Both bootstrap methods are fitted at $L = 10, 20, 40$. The primary choice is $L = 20$, carried forward from Parts B–C: it permits within-block pairs at every selected lag while retaining approximately 50 fixed blocks. Lengths 10 and 40 assess how the conclusion changes when less or more temporal dependence is preserved.

Because this application involves only one observed dataset, we increase the bootstrap budget to $B = 500$ and repeat each method–length setting with five independent bootstrap seeds. These 30 fits assess Monte Carlo stability conditional on the same observations; they do not constitute 30 independent datasets or provide a sampling confidence interval for the true dimension. For each fit, we examine the full curves f, ϕ and g, the selected dimension, and the gap between the smallest and second-smallest values of g.

Estimated dimension and supporting evidence

Table 4: Dimension estimates across five bootstrap seeds, with $B = 500$ per fit. The last column is the smallest gap between the two lowest criterion values across those five runs.

Method	L	Observed $\hat{q}$	Selection counts	Minimum gap
Fixed	10	2	2: 5 runs	0.02019
Stationary	10	2 or 3	2: 4; 3: 1	0.00069
Fixed	20	3	3: 5 runs	0.01278
Stationary	20	3	3: 5 runs	0.02273
Fixed	40	3	3: 5 runs	0.02783
Stationary	40	3	3: 5 runs	0.03459

Both methods consistently select $\hat{q} = 3$ at $L = 20$ and $L = 40$ (Table 4). For the primary fixed-block fit with $L = 20$ and the first seed,

$$f(3) = 0.04660, \qquad \phi(3) = 0.01649, \qquad g(3) = 0.06310.$$

The next-best candidate is $k = 2$, whose criterion is larger by 0.01278. The first four ordered component strengths are approximately 0.39989, 0.34474, 0.14143 and 0.03383. The drop after the third direction supports a three-dimensional signal subspace, while the bootstrap component penalizes the instability introduced by additional weak directions.

Figure 7 shows how the two contributions determine the minimum. The scree contribution is shared by the bootstrap settings because it depends on the original sample, whereas f changes with the resampling method and block length. At $L = 20$ and 40, the combination has its minimum at 3 for both methods. However, the results are not uniform over all block lengths: fixed blocks of length 10 select 2 in every run, and stationary blocks with expected length 10 select 2 in four runs and 3 in one. The particularly small gap for the latter setting indicates that the two candidates are close; it is not a hypothesis-test p-value or a confidence level.

Investigating block-length sensitivity

Figure 8 shows that the main discrepancy concerns the competition between $k = 2$ and $k = 3$. To investigate it, we examine the ordered components obtained from the original-data decomposition,

$$\widehat{Z} = (X - \mathbf{1}_n \bar{x}^\mathsf{T})\widehat{W}\widehat{U}.$$

The third component has its largest absolute sample autocorrelation over lags 1–12 at lag $h = 8$, where its value is approximately 0.287. This relatively long-lag feature is especially vulnerable to short blocks.

For fixed blocks, the approximate fraction of lag-h pairs lying within one block is $\max(1 - h/L, 0)$, ignoring finite-series end effects. For stationary blocks, the probability of no restart over h steps is

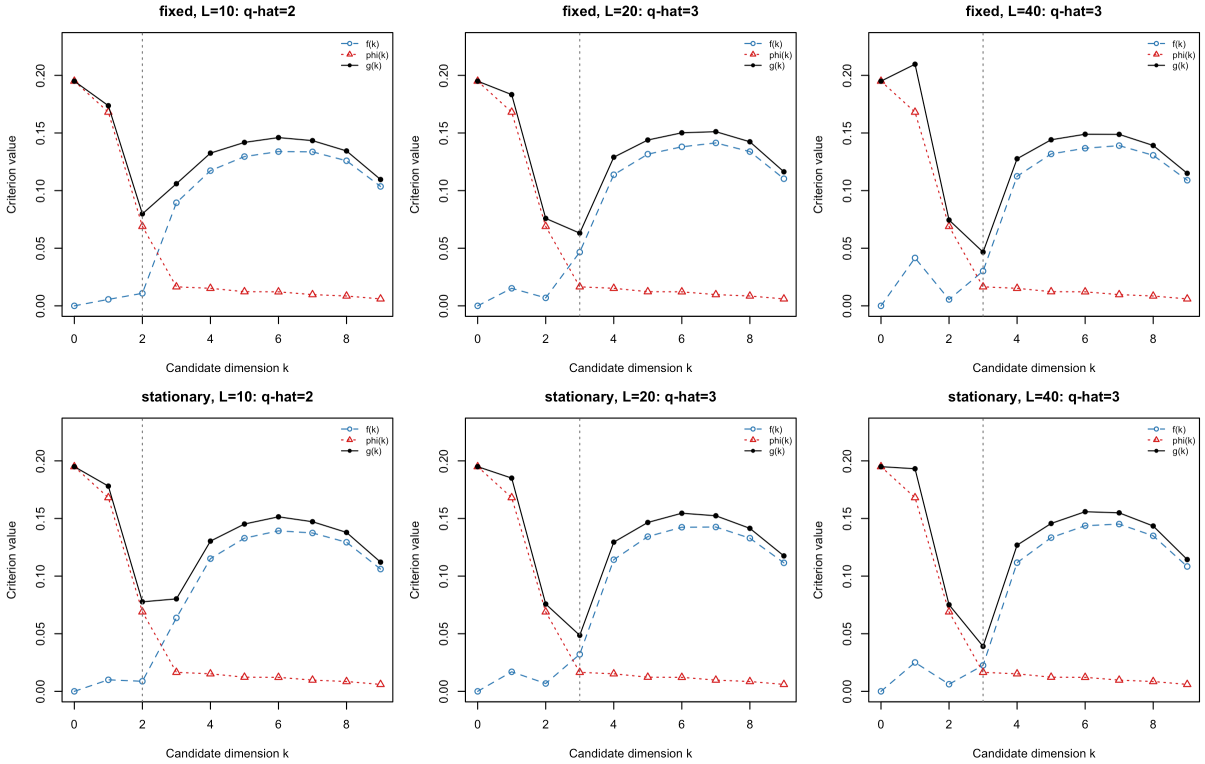


Figure 7: Ladle components for all six settings on the course data, using the first seed in each setting. The vertical line marks the minimizing dimension. The value $\phi(k)$ corresponds to the strength of the $(k+1)$st direction, as in Part C.

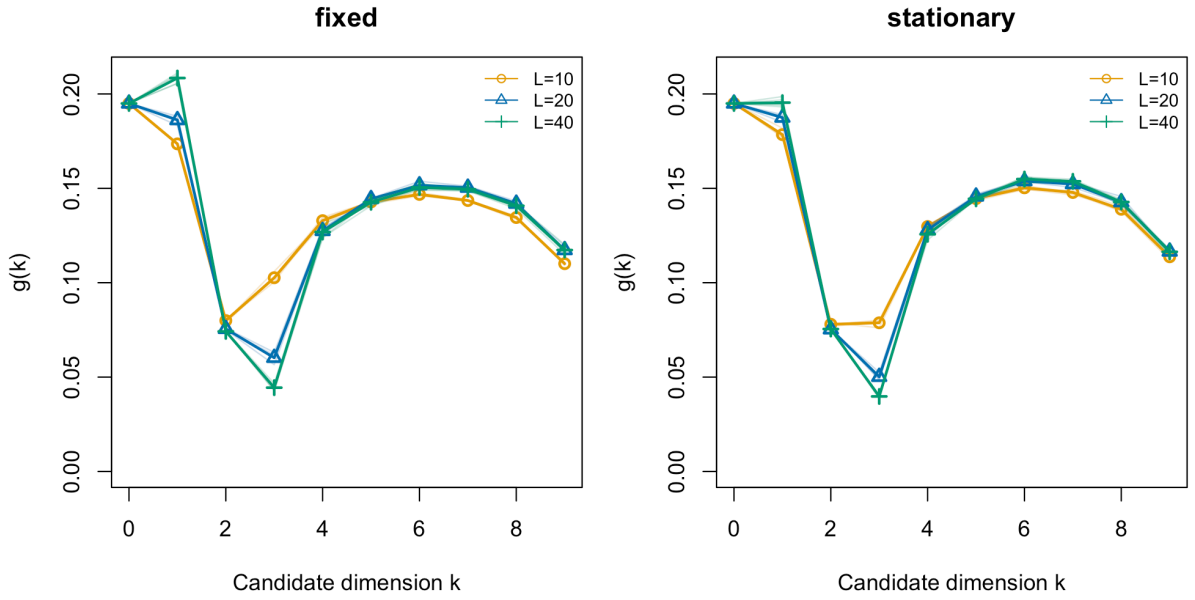


Figure 8: Sensitivity of $g(k)$ to block length. Faint curves show the five individual seed runs; the heavier curves show their means. The reported dimension estimates are calculated separately for each run.

$(1 - 1/L)^h$. At $h = 8$, these quantities are

L	10	20	40
Fixed: within-block fraction	0.200	0.600	0.800
Stationary: no-restart probability	0.430	0.663	0.817

They describe preservation of local ordering, not exact autocorrelation attenuation factors or dimension-recovery probabilities. Nevertheless, they provide a plausible explanation for the observed sensitivity: short blocks weaken the dependence supporting the third direction, increasing the bootstrap instability of the three-dimensional subspace. Increasing L makes the minimum at $k = 3$ clearer. The agreement at $L = 20$ and 40, together with this dependence pattern, supports reporting $\hat{q} = 3$ as the main conclusion while explicitly retaining the short-block alternative.

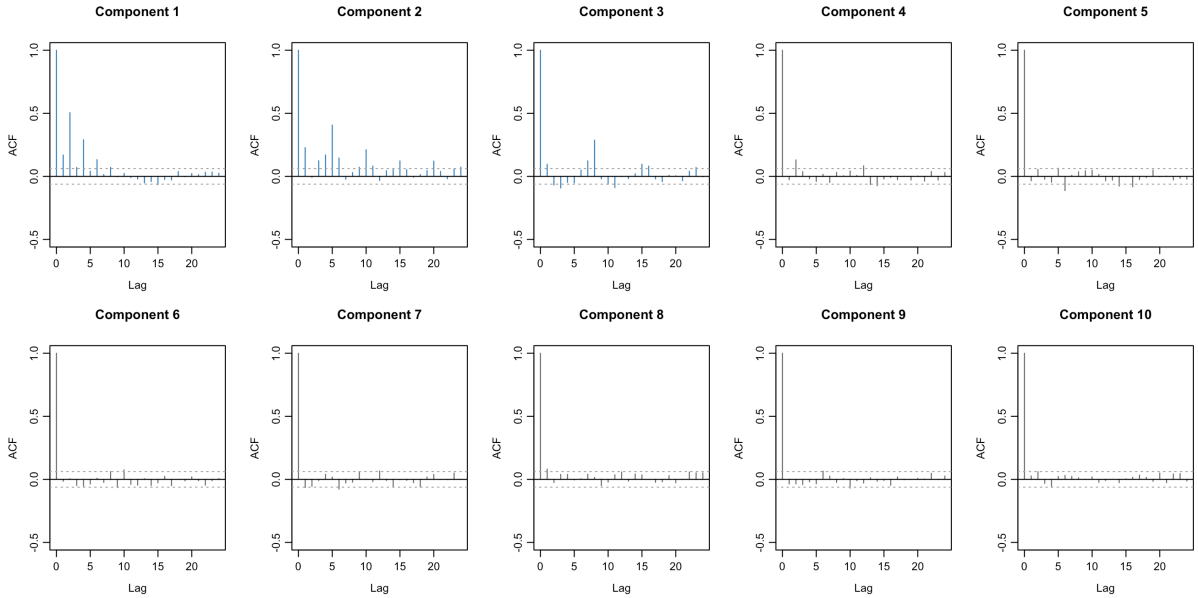


Figure 9: Sample ACFs of the ordered components. The first three directions, classified as signals by the primary fit, are blue. Dashed lines at $\pm 1.96/\sqrt{1000}$ are descriptive white-noise reference bands, not simultaneous confidence bounds after estimating the directions.

Interpretation and limitations

Under the second-order source-separation model, the main result suggests three serially dependent linear combinations of the ten observed variables and seven remaining directions with weaker dependence at the selected lags. The first three component ACFs have pronounced features at lags 2, 5 and 8, respectively (Figure 9). The remaining component strengths are smaller, although their sample ACFs are not identically zero.

The estimated dimension counts latent linear combinations, not a subset of three original variables. Nor does the classification establish that the remaining directions are independent, Gaussian, or free of nonlinear dependence. The interpretation is conditional on approximate stationarity, the SOS model and the selected lag set. Finally, successful recovery in Setting 1 validates the computational procedure under

that design; it does not establish the unknown dimension of this particular dataset. Here the evidence for three directions comes from the criterion curves, the longer-block agreement and the observed dependence structure, with the sensitivity at $L = 10$ reported as part of the result.

Task 2: Bayesian mixture modelling with MCMC

Part A: Latent-variable representation and Gibbs sampler

Part B: Simulation study and computational behaviour

Part C: Label switching and identifiability

Part D: Application to real data